

Drone Payload Structure Design and Analysis

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Task 1: Gather your starter assumptions and load material properties

Load the provided `droneArmMaterials.mat` file into MATLAB. This contains a data struct with the material properties for six different materials commonly used for drones (carbon fiber composite, aluminum alloy, fiberglass composite, PLA plastic, ABS plastic, and wood). For each material, you have been provided the following information:

- Density (ρ), in $\frac{\text{kg}}{\text{m}^3}$
- Young's Modulus (E) in Pa
- Poisson's ratio (ν), unitless
- Yield strength, in Pa
- Average cost per meter in US dollars

For this project, you can assume the following design parameters:

- Quadcopter drone with 4 motors and 4 propellers
- Each motor can provide up to 1kg thrust
- Acceleration due to gravity: $g = 9.81 \frac{\text{m}}{\text{s}^2}$
- Total weight of the drone components without the arms or the payload: 1kg

The breakdown of the weight of each drone component is as follow:

- Battery = 0.3kg
- Each motor = 65g, 4 motors = 0.26kg
- Each propeller = 10g, 4 propellers = 0.04 kg
- Each electronic speed controller = 30g, 4 electronic speed controllers = 0.12 kg

- Flight controller and electronics = 0.05kg
- Receiver/GPS = 0.03kg
- Drone frame (center body) excluding arms = 0.2kg

```
clear; clc;

load('droneArmMaterials.mat');

nMaterials = numel(materials);

fprintf('Loaded %d candidate materials:\n\n', nMaterials);
```

Loaded 6 candidate materials:

```
fprintf('%-20s %12s %12s %8s %16s %14s\n', ...
    'Material', 'rho (kg/m3)', 'E (Pa)', 'nu', 'Yield (Pa)', 'Cost ($/m)');
```

Material	rho (kg/m3)	E (Pa)	nu	Yield (Pa)	Cost (\$/m)
----------	-------------	--------	----	------------	-------------

```
fprintf('%s\n', repmat('-', 1, 90));
```

```
for i = 1:nMaterials
    fprintf('%-20s %12.1f %12.3e %8.3f %16.3e %14.2f\n', ...
        materials(i).name, ...
        materials(i).rho_kg_m3, ...
        materials(i).E_Pa, ...
        materials(i).nu, ...
        materials(i).yieldStrength_Pa, ...
        materials(i).cost_USD_per_m);
end
```

Carbon Fiber Composite (CFRP)	1600.0	7.000e+10	0.300	6.000e+08	30.00
Aluminum Alloy	2700.0	6.900e+10	0.330	2.750e+08	7.50
Fiberglass Composite (GFRP)	1900.0	2.500e+10	0.280	3.000e+08	15.00
PLA Plastic	1250.0	3.500e+09	0.360	6.000e+07	3.50
ABS Plastic	1050.0	2.000e+09	0.350	4.000e+07	3.50
Wood (Birch)	600.0	1.000e+10	0.350	8.000e+07	2.00

```
fprintf('\n');

g = 9.81;

numMotors = 4;
numArms = 4;

thrustPerMotor_kg = 1;
thrustPerMotor_N = thrustPerMotor_kg * g;
```

```
totalThrust_N      = numMotors * thrustPerMotor_N;
```

```
baseMass_kg = 1;
```

```
minPayload_kg = 0.5;
```

```
minTWratio      = 2;
```

```
fprintf('Design Assumptions:\n');
```

Design Assumptions:

```
fprintf('  Number of motors/arms: %d\n', numMotors);
```

Number of motors/arms: 4

```
fprintf('  Max thrust per motor: %.2f kgf (%.2f N)\n', thrustPerMotor_kg,  
thrustPerMotor_N);
```

Max thrust per motor: 1.00 kgf (9.81 N)

```
fprintf('  Total max thrust (4 motors): %.2f N\n', totalThrust_N);
```

Total max thrust (4 motors): 39.24 N

```
fprintf('  Base drone mass (no arms/payload): %.2f kg\n', baseMass_kg);
```

Base drone mass (no arms/payload): 1.00 kg

```
fprintf('  Minimum required payload: %.2f kg\n', minPayload_kg);
```

Minimum required payload: 0.50 kg

```
fprintf('  Minimum thrust-to-weight ratio: %.1f : 1\n', minTWratio);
```

Minimum thrust-to-weight ratio: 2.0 : 1

```
fprintf('  Gravity: %.2f m/s^2\n\n', g);
```

Gravity: 9.81 m/s²

Task 2: Propose at least two drone arm designs

Develop at least two distinct drone arm designs by drawing clear, neatly labeled sketches that show key geometric features, including:

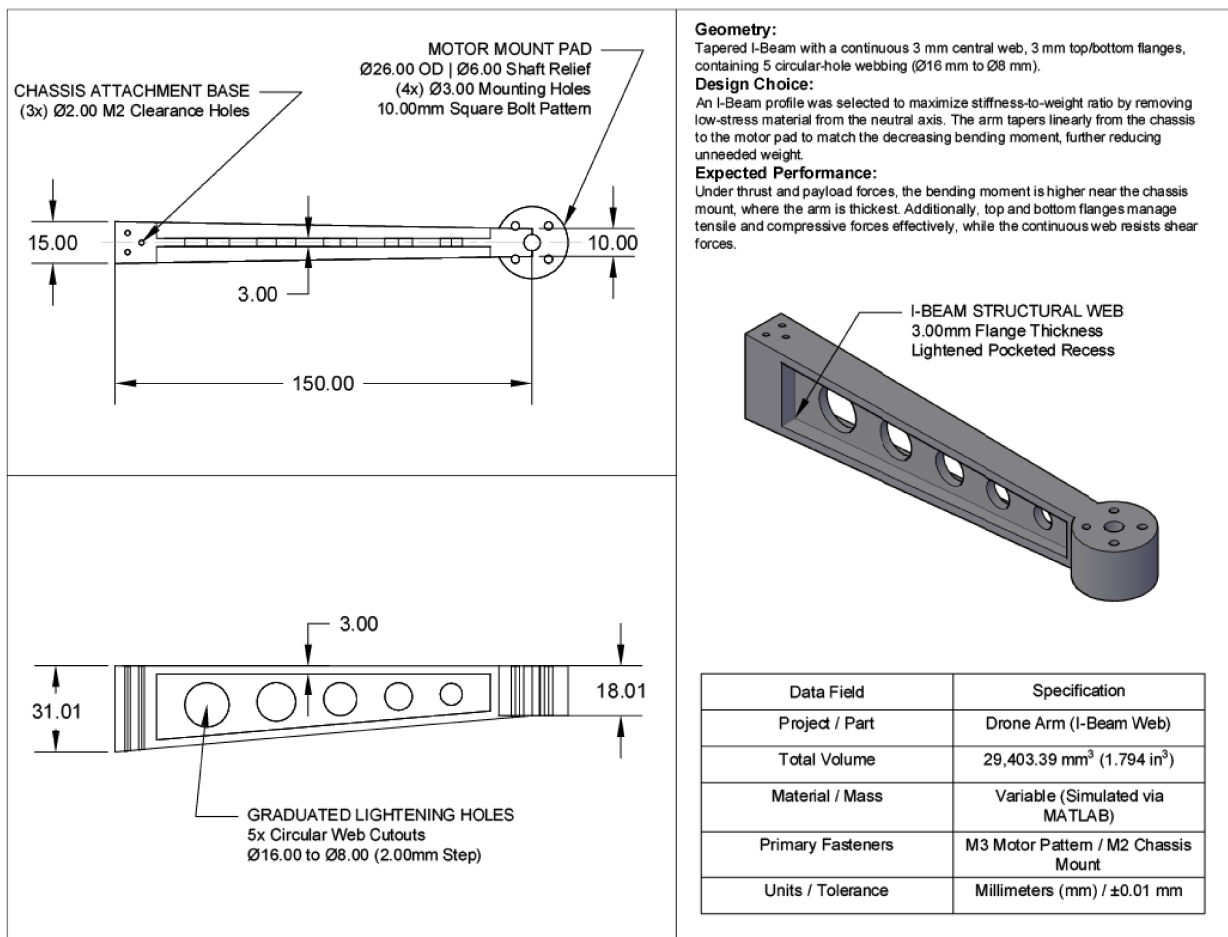
- Overall shape (e.g. beam, tube, truss structure)
- Dimensions (length, width, thickness)
- Motor mount location
- Where the arm connects to the drone body

The two designs should be meaningfully different from each other (e.g. different geometries or structural strategies) and not just small variations of the same idea.

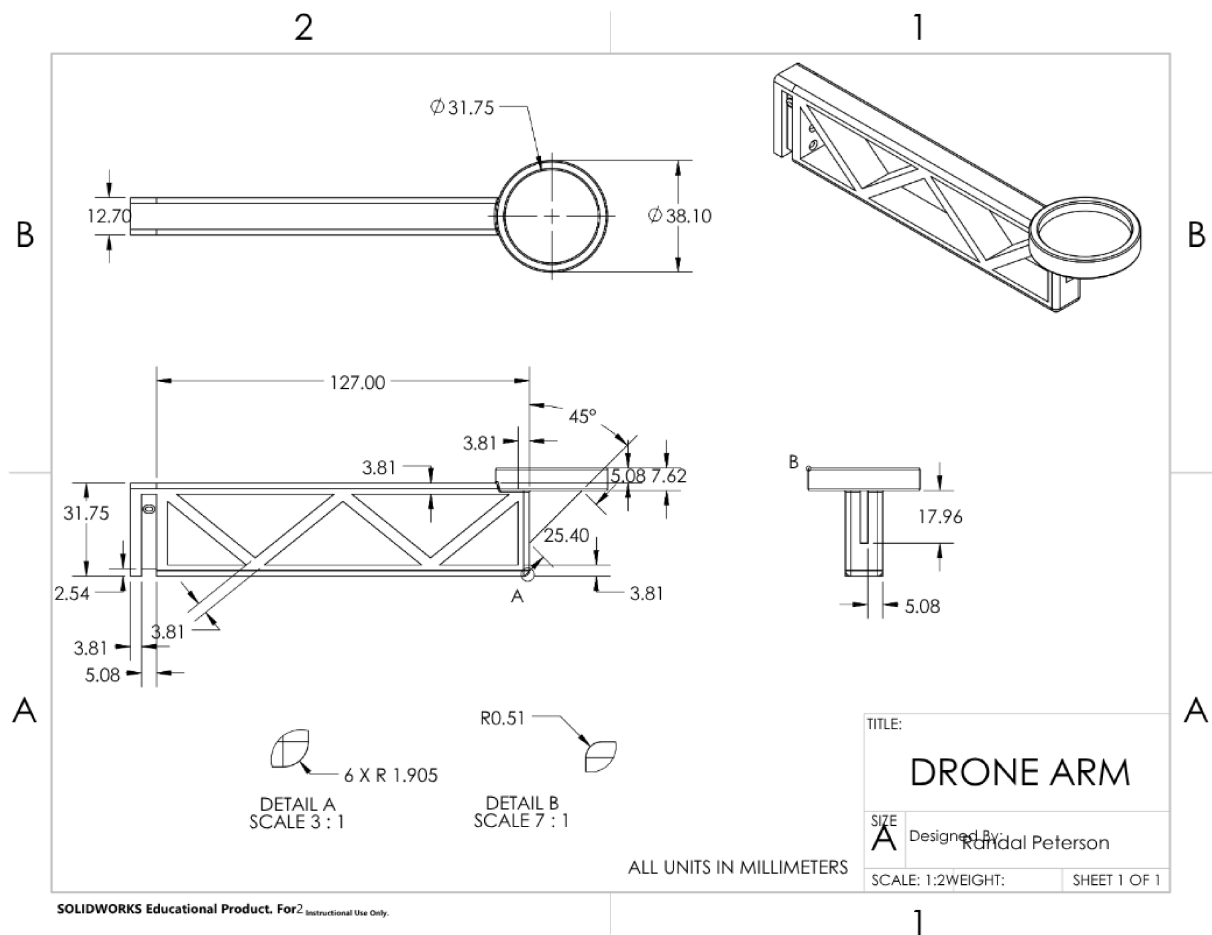
Navigate to the 'Insert' tab and insert pictures of your sketches in the space below (hand-drawn or digital). Include a short design description for each sketch that includes:

- Geometry (e.g. solid beam, hollow tube, beam with trusses)
 - Why you chose this design
 - How it might perform under load
-
- Design 1 (I-Beam + Holes):
 - **Geometry:** Tapered I-Beam with a continuous 3 mm central web, 3 mm top/bottom flanges, containing 5 circular-hole webbing ($\varnothing 16$ mm to $\varnothing 8$ mm).
 - **Design Choice:** An I-Beam profile was selected to maximize stiffness-to-weight ratio by removing low-stress material from the neutral axis. The arm tapers linearly from the chassis to the motor pad to match the decreasing bending moment, further reducing unneeded weight.
 - **Expected Performance:** Under thrust and payload forces, the bending moment is higher near the chassis mount, where the arm is thickest. Additionally, top and bottom flanges manage tensile and compressive forces effectively, while the continuous web resists shear forces.
-
- Design 2 (Truss):
 - The second drone arm is designed based on trusses. It involves a straight lower and upper beam connected with trusses from the front until the end of the arm. It involves a shallow circular slot for the motor to sit on and for the propeller. It also involves a rib below the slot for more stability for the motor mount. To connect to the drone body, a slot to slide in and hold the arm, along with two screw holes to lock down the arm. This design was chosen based on bridge designs. Many bridges and other structures use trusses to distribute the load, so it was assumed that it would hold up well under load. Under load, it is assumed to perform well and hold its shape from the thrust and weight of the motor.

```
img = imread('holes.png');  
rotatedImg = imrotate(img, 90);  
imshow(rotatedImg);
```



```
img = imread('truss.png');
rotatedImg = imrotate(img, 0);
imshow(rotatedImg);
```



Task 3: Perform thrust-to-weight analysis across all design and material options

Write a MATLAB function that will perform a thrust-to-weight analysis across all design and material options. This analysis will help you evaluate which drone arm design and which material optimizes payload capacity while meeting suggested safety requirements.

- Your design must support a minimum payload of 0.5kg.
- Your design must meet a thrust-to-weight ratio of at least 2:1 to ensure safe and stable flight. This safety standard is widely used in drone design. Conceptually, it means that the drone's maximum thrust must be at least twice its weight. This ensures stable hovering and maneuvering with a safety margin. Note that the thrust-to-weight safety margin is a system-level safety standard for drone flight that is distinct from the structural safety factor of the drone arms (which you will evaluate later through finite element analysis).

- Your goal is calculate maximum payload capacity for each design and material option (i.e. what is the heaviest payload your drone design could carry while still allowing the drone to safely take off?)

Ensure that the results of your analysis can be readily interpreted, e.g. in the form of a summary table or plot.

Note: if you have two drone arm designs, the expected output should be 12 payload capacity estimates (2 designs across 6 materials options).

```
clear; clc;

%% 1. Load material properties
load('droneArmMaterials.mat'); % loads variable "materials" (1x6 struct array)

nMaterials = numel(materials);

getName     = @(i) string(materials(i).name);
getDensity  = @(i) materials(i).rho_kg_m3;
getCost     = @(i) materials(i).cost_USD_per_m;

fprintf('Loaded %d materials:\n', nMaterials);
```

Loaded 6 materials:

```
for i = 1:nMaterials
    fprintf(' %d. %s (rho = %.1f kg/m^3)\n', i, getName(i), getDensity(i));
end
```

1. Carbon Fiber Composite (CFRP) (rho = 1600.0 kg/m³)
2. Aluminum Alloy (rho = 2700.0 kg/m³)
3. Fiberglass Composite (GFRP) (rho = 1900.0 kg/m³)
4. PLA Plastic (rho = 1250.0 kg/m³)
5. ABS Plastic (rho = 1050.0 kg/m³)
6. Wood (Birch) (rho = 600.0 kg/m³)

```
fprintf('\n');

%% 2. parameter
g = 9.81;
numMotors = 4;
numArms = 4;
thrustPerMotor_kg = 1;
thrustPerMotor_N = thrustPerMotor_kg * g;
totalThrust_N = numMotors * thrustPerMotor_N;

baseMass_kg = 1;
minPayload_kg = 0.5;
minTWratio = 2;
```

```

%% 3. Arm geometry

in3_to_m3 = 1.6387e-5;

armVolume_Design1_truss = 1.81 * in3_to_m3;
armVolume_Design2_holes = 1.79 * in3_to_m3;

designNames = {'Design 1: Truss/Lattice Arm', 'Design 2: Solid Arm w/
Holes'};
armVolumes = [armVolume_Design1_truss, armVolume_Design2_holes];
nDesigns = numel(designNames);

%% 4. Run thrust-to-weight analysis for every design x material combo
Design      = strings(nDesigns * nMaterials, 1);
Material     = strings(nDesigns * nMaterials, 1);
ArmMass_kg   = zeros(nDesigns * nMaterials, 1);
TotalMass_kg = zeros(nDesigns * nMaterials, 1);
TWratio      = zeros(nDesigns * nMaterials, 1);
MaxPayload_kg = zeros(nDesigns * nMaterials, 1);
MeetsMin2to1 = false(nDesigns * nMaterials, 1);
MeetsMinPayload = false(nDesigns * nMaterials, 1);

row = 1;
for d = 1:nDesigns
    for m = 1:nMaterials
        density = getDensity(m);

        [maxPayload, tw, totalMass] = thrustToWeightAnalysis( ...
            armVolumes(d), density, numArms, baseMass_kg, ...
            totalThrust_N, minTWratio, g);

        Design(row)      = string(designNames{d});
        Material(row)     = getName(m);
        ArmMass_kg(row)   = numArms * armVolumes(d) * density;
        TotalMass_kg(row) = totalMass;
        TWratio(row)      = tw;
        MaxPayload_kg(row) = maxPayload;
        MeetsMin2to1(row) = tw >= minTWratio;
        MeetsMinPayload(row) = maxPayload >= minPayload_kg;

        row = row + 1;
    end
end

results = table(Design, Material, ArmMass_kg, TotalMass_kg, TWratio, ...
    MaxPayload_kg, MeetsMin2to1, MeetsMinPayload);

%% 5. Display results, sorted by max payload capacity
results = sortrows(results, 'MaxPayload_kg', 'descend');
disp(results);

```

Design	Material	ArmMass_kg	TotalMass_kg	TW
"Design 2: Solid Arm w/ Holes"	"Wood (Birch)"	0.070399	1.0704	3.7
"Design 1: Truss/Lattice Arm"	"Wood (Birch)"	0.071185	1.0712	3.7
"Design 2: Solid Arm w/ Holes"	"ABS Plastic"	0.1232	1.1232	3.5
"Design 1: Truss/Lattice Arm"	"ABS Plastic"	0.12457	1.1246	3.5
"Design 2: Solid Arm w/ Holes"	"PLA Plastic"	0.14666	1.1467	3.4
"Design 1: Truss/Lattice Arm"	"PLA Plastic"	0.1483	1.1483	3.4
"Design 2: Solid Arm w/ Holes"	"Carbon Fiber Composite (CFRP)"	0.18773	1.1877	3.3
"Design 1: Truss/Lattice Arm"	"Carbon Fiber Composite (CFRP)"	0.18983	1.1898	3.3
"Design 2: Solid Arm w/ Holes"	"Fiberglass Composite (GFRP)"	0.22293	1.2229	3.2
"Design 1: Truss/Lattice Arm"	"Fiberglass Composite (GFRP)"	0.22542	1.2254	3.2
"Design 2: Solid Arm w/ Holes"	"Aluminum Alloy"	0.31679	1.3168	3.0
"Design 1: Truss/Lattice Arm"	"Aluminum Alloy"	0.32033	1.3203	3.0

```
% Identify best combination
validRows = results(results.MeetsMin2to1 & results.MeetsMinPayload, :);
if ~isempty(validRows)
    best = validRows(1, :);
    fprintf('\nBest design/material combo: %s with %s\n', best.Design,
best.Material);
    fprintf('Max payload: %.3f kg | T/W ratio: %.2f\n', best.MaxPayload_kg,
best.TWratio);
else
    fprintf('\nNo combination meets both the 2:1 T/W ratio and 0.5 kg
payload requirement.\n');
end
```

Best design/material combo: Design 2: Solid Arm w/ Holes with Wood (Birch)
Max payload: 0.930 kg | T/W ratio: 3.74

```
function [maxPayload_kg, TWratio, totalMass_kg] = thrustToWeightAnalysis(
...
    armVolume, density, numArms, baseMass_kg, totalThrust_N, minTWratio, g)
% Computes arm mass, and solves for the max payload so that the
% thrust to weight ratio is exactly at the minimum required threshold

armMass_kg = numArms * armVolume * density;
totalMass_kg = baseMass_kg + armMass_kg;

% Max total takeoff weight allowed
maxTotalWeight_N = totalThrust_N / minTWratio;
maxTotalMass_kg = maxTotalWeight_N / g;

maxPayload_kg = maxTotalMass_kg - totalMass_kg;

% T/W ratio
TWratio = totalThrust_N / (totalMass_kg * g);
end
```

Task 4: Perform finite element analysis across all design and material options

Perform finite element analysis (FEA) across all design and material options to evaluate the structural safety factor of your drone arm design. To simplify the task, you will perform the FEA only on a single drone arm.

Using a CAD software of your choice, first create a 3D model of one drone arm for each of your designs. Be sure your model can be exported as a STEP or STL file for analysis in MATLAB.

- Recommended CAD software options: Tinkercad, Onshape, Fusion 360, Solidworks (most have free versions)

Import your model into MATLAB and perform FEA. In your FEA model, you should model and apply two loads to the drone arm: upward thrust force from the motor and downward weight of the motor.

- Helpful resources: [online tutorial](#), documentation for [Structural Mechanics](#) and [Partial Differentiation Equation Toolbox](#), and [example workflow](#)

For each design and material combination, you should report the overall maximum displacement, maximum stress, and factor of safety and include visualizations of the x-, y-, and z-displacement and Von Mises stress. Ensure that your results are presented for easy interpretation (e.g. present a table with numerical results, clearly labeled subplots for the visualizations).

- The factor of safety compares how strong a material is to how much stress it experiences, indicating whether a design is safe under the given loads. For this project use the equation:

$$FoS = \frac{\text{Material Yield Strength (Pa)}}{\text{Maximum von Mises Stress}}$$

Yield strength indicates how much stress a given material can withstand before permanently deforming. Maximum von Mises stress (derived from your FEA model) is a measure of how much stress your design experiences under the applied loads.

- A typical engineering safety margin for FoS is ~1.5-2+
- This [example](#) demonstrates the plots you are asked to include in your results

%This test was done on both designs using all 6 of our proposed materials to provide our team with enough data to determine which arm design performed better. The conditions that were held constant throughout every calculation were the thrust produced by the drones propellers (4kg) and the weight of both the drone body and minimum payload (1kg & 0.5kg). The conditions that were dependent on the density of its material was the weight of the drone arm and the additional possible payload weight.

%The final determination of which drone arm design performed best was made from calculating the Displacement, Von Mises Stress, and Factor of safety.

These calculations were simplified using the functions provided in the PDE toolbox.

Drone Arm 1 FEA

```
clear
clc

%----- Load in Important Data / Create Mesh / Known Values -----%
prop = load('droneArmMaterials.mat');

% Material Number (n)
% n=1, Carbon Fiber Composite
% n=2, Aluminum Alloy
% n=3, Fiberglass Composite
% n=4, PLA Plastic
% n=5, ABS Plastic
% n=6, Wood (Birch)

% Import CAD file
gm = importGeometry("DRONE ARM.STL"); %Note: CAD model is flipped on it's
side (+y is up)
```

Error using importGeometry (line 34)
Invalid argument at position 1. File DRONE ARM.STL does not exist.

```
gm = scale(gm,0.001); %Scaled down geometry to keep units
in (m)
%pdegplot(gm,FaceLabels="on",FaceAlpha=0.7) % <---Use to visualize
Geometry (not needed)

% Create mesh of CAD file
md = femodel(AnalysisType="structuralStatic",Geometry=gm);
md = generateMesh(md, Hmin=0.001);
%pdemesh(md) % <---Use to visualize mesh (not needed)

% Known/Set Values
n = 1; %Material Number
TotalA = prop.materials(n).rho_kg_m3*0.00002966*-9.8; %Arm Weight, F(g) =
p*V*a(g) (N)
TotalT = 1 * 9.8; %Thrust force, F(t)
= m*a(+g) (N)
TotalW = 1 * -9.8; %Body Weight, F(g)
= m*a(g) (N)
TotalP = 0.5 * -9.8; %Min Payload
Weight, F(g) = m*a(g) (N)
AreaProp = pi*(0.03175/2)^2; %Area of propeller
mount (m^2)
AreaBottom = 0.127*0.0127; %Area underneath
Drone Arm (m^2)
```

```

%----- Set Conditions -----%
% Add in Material properties onto Mesh
md.MaterialProperties = materialProperties(...
YoungsModulus=prop.materials(n).E_Pa,...
MassDensity=prop.materials(n).rho_kg_m3,...
PoissonsRatio=prop.materials(n).nu);

% LOADS (aka conditions):
% Add in constraint (End of Drone Arm)
md.FaceBC(39) = faceBC(Constraint="fixed");

% Factor in weight of drone arm
md.CellLoad = cellLoad(Gravity=[0 -9.8 0]); % Gravity applied in y instead
of z

% Factor in thrust force of propeller (1kg per prop)
TPressure = TotalT/AreaProp; %Pressure of Thrust
md.FaceLoad(36) = faceLoad(SurfaceTraction = [0 TPressure 0]);

% Factor in weight of drone components in body (1kg)
W0nArm = TotalW/4; %Assuming center of mass is at middle (N) and weight is
divided equally per arm
WPressure = W0nArm/AreaBottom;
md.FaceLoad(15) = faceLoad(SurfaceTraction = [0 WPressure 0]);

% Factor in Payload weight (min of 0.5kg)
P0nArm = TotalP/4;
PPressure = P0nArm/AreaBottom;
md.FaceLoad(15) = faceLoad(SurfaceTraction = [0 PPressure 0]);

% Factor in Additional Payload while keeping thrust to weight ratio of 2:1
AddP = -1*(((4*TotalT)/2)+TotalW+TotalP+(TotalA*4));
TotalAP = AddP/4;
APPressure = TotalAP/AreaBottom;
md.FaceLoad(15) = faceLoad(SurfaceTraction = [0 APPressure 0]);

%---- Calculate FEA Solution / Analyze Solutions / Visual -----%
% Solve FEA
sol = solve(md);

```

```

% Calculating max displacement (in y direction)
maxDeflection = max(sol.Displacement.uy)

% Calculating max Von Mises Stress
maxVonStress = max(sol.VonMisesStress)

% Factor of Safety (How close to plastic deformation (yield strength))
FoS = prop.materials(n).yieldStrength_Pa/maxVonStress


% Visual of Results (x,y,z displacements and Von Mises Stress)
% Data to visualize
meshData = sol.Mesh;
nodalData = sol.Displacement.Magnitude;
deformationData = sol.Displacement;

% Create PDE result visualization
resultViz = pdeviz(meshData,abs(real(nodalData)), ...
    "DeformationData",deformationData, ...
    "DeformationScaleFactor",457.5338, ...
    "ColorLimits",[0 8.05e-05]);

% Clear temporary variables
clearvars meshData nodalData deformationData

```

Drone Arm 2 FEA

```

clear
clc

%----- Load in Important Data / Create Mesh / Known Values -----%
prop = load('droneArmMaterials.mat');

% Material Number (n)
% n=1, Carbon Fiber Composite
% n=2, Aluminum Alloy
% n=3, Fiberglass Composite
% n=4, PLA Plastic
% n=5, ABS Plastic
% n=6, Wood (Birch)

% Import CAD file
gm = importGeometry("drone_circles_arm1.stl"); %Note: CAD model is flipped
on it's side (+y is up)
gm = scale(gm,0.001); %Scaled down geometry to
keep units in (m)

```

```

pdegplot(gm,FaceLabels="on",FaceAlpha=0.7)    % <---Use to visualize
Geometry (not needed)

% Create mesh of CAD file
md = femodel(AnalysisType="structuralStatic",Geometry=gm);
md = generateMesh(md, Hmin=0.001);
%pdemesh(md)    % <---Use to visualize mesh (not needed)

% Known/Set Values
n = 1;
TotalA = prop.materials(n).rho_kg_m3*0.00002933*-9.8;    %Material Number
p*V*a(g) (N)    %Arm Weight, F(g) =
TotalT = 1 * 9.8;    %Thrust force, F(t)
= m*a(+g) (N)
TotalW = 1 * -9.8;    %Body Weight, F(g)
= m*a(g) (N)
TotalP = 0.5 * -9.8;    %Min Payload
Weight, F(g) = m*a(g) (N)
AreaProp = pi*(0.026/2)^2;    %Area of propeller
mount (m^2)
AreaBottom = 0.150*0.0150;    %Area underneath
Drone Arm (m^2)

%----- Set Conditions -----%
% Add in Material properties onto Mesh
md.MaterialProperties = materialProperties(...
YoungsModulus=prop.materials(n).E_Pa,...
MassDensity=prop.materials(n).rho_kg_m3,...
PoissonsRatio=prop.materials(n).nu);

% LOADS (aka conditions):
% Add in constraint (End of Drone Arm)
md.FaceBC(29) = faceBC(Constraint="fixed");

% Factor in weight of drone arm
md.CellLoad = cellLoad(Gravity=[0 -9.8 0]); % Gravity applied in y instead
of z

% Factor in thrust force of propeller (1kg per prop)
TPressure = TotalT/AreaProp; %Pressure of Thrust
md.FaceLoad(30) = faceLoad(SurfaceTraction = [0 TPressure 0]);

% Factor in weight of drone components in body (1kg)

```

```

W0nArm = TotalW/4; %Assuming center of mass is at middle (N) and weight is
divided equally per arm
WPressure = W0nArm/AreaBottom;
md.FaceLoad(11) = faceLoad(SurfaceTraction = [0 WPressure 0]);

% Factor in Payload weight (min of 0.5kg)
P0nArm = TotalP/4;
PPressure = P0nArm/AreaBottom;
md.FaceLoad(11) = faceLoad(SurfaceTraction = [0 PPressure 0]);

% Factor in Additional Payload while keeping thrust to weight ratio of 2:1
AddP = -1*((4*TotalT)/2)+TotalW+TotalP+(TotalA*4);
TotalAP = AddP/4;
APPressure = TotalAP/AreaBottom;
md.FaceLoad(11) = faceLoad(SurfaceTraction = [0 APPressure 0]);

%---- Calculate FEA Solution / Analyze Solutions / Visual -----%
% Solve FEA
sol = solve(md);

% Calculating max displacement (in y direction)
maxDeflection = max(sol.Displacement.uy)

% Calculating max Von Mises Stress
maxVonStress = max(sol.VonMisesStress)

% Factor of Safety (How close to plastic deformation (yield strength))
FoS = prop.materials(n).yieldStrength_Pa/maxVonStress

% Visual of Results (x,y,z displacements and Von Mises Stress)
% Data to visualize
meshData = sol.Mesh;
nodalData = sol.Displacement.Magnitude;
deformationData = sol.Displacement;

% Create PDE result visualization
resultViz = pdeviz(meshData,abs(real(nodalData)), ...
    "DeformationData",deformationData, ...
    "DeformationScaleFactor",457.5338, ...
    "ColorLimits",[0 8.05e-05]);

% Clear temporary variables
clearvars meshData nodalData deformationData

%final results for FEA

```

% Drone #1 (design with holes)		
% Material	Max Displacement	Max Von Mises
Stress Factor of Safety		
% Carbon Fiber Composite	1.14E-05	3.77E+06
159.0232		
% Aluminum Alloy	1.14E-05	3.71E+06
74.0742		
% Fiberglass Composite	3.18E-05	3.79E+06
79.1552		
% PLA Plastic	2.26E-04	3.71E+06
16.1702		
% ABS Plastic	3.97E-04	3.73E+06
10.7333		
% Wood (Birch)	7.96E-05	3.74E+06
21.408		
% Drone #2 (Truss design)		
% Carbon Fiber Composite	8.66E-06	1.18E+06
507.1589		
% Aluminum Alloy	8.70E-06	1.16E+06
236.1791		
% Fiberglass Composite	2.42E-05	1.19E+06
251.9274		
% PLA Plastic	1.74E-04	1.17E+06
51.1028		
% ABS Plastic	3.04E-04	1.18E+06
33.9911		
% Wood (Birch)	6.11E-05	1.18E+06
67.7559		

Interpretation of Results - Final Recommended Design Solution

Using the results of your thrust-to-weight analysis and finite element analysis, recommend a final drone arm design that meets all required constraints. Your recommendation should clearly identify the selected design (arm geometry and material choice), make use of quantitative results to justify your decisions, demonstrate that the design meets the constraints, explain the trade-offs considered, and include comparisons to alternative designs.

- In our Thrust to Weight analysis, our team was able to realize that our Drone Arm Design #2 performed the best at maximizing payload for every material type. It was also noted that when it was constructed out of birch wood, it outperformed every other option. It was up to 1.5 times as much as our lowest option which was using the Aluminum Alloy.
- When considering our FEA results, our team noted again that our Drone Arm design #2 was once again outperforming Design #1 for every material type according to our factor of safety.
- We had to make a note that our previous outperformer, Birch Wood, was no longer performing better than our other material options. In this analysis, Carbon Fiber was now leading in the material type category due to its higher factor of safety.
- Factoring in both tests, we were able to come up with our final decision. We believe that our Drone Arm Design #2 performed best out of our two designs due to its slim tapering shape and it is paired best when it is built out of Carbon Fiber Composite. It is the best middle ground when considering its payload capacity and highest performing when considering the materials strength capabilities.

What are some limitations of your work?

- Limited by FEA assumptions during boundary conditions (placement of drone body weight and payload weight)
- Limited by the number of designs we tested. Testing more geometries would have resulted in a more comprehensive and dependable result.

What are practical next steps?

- Based on the results of both designs, a practical next step is to take the best aspect of each design and combine them into a refined design. The geometry is key to creating an effective arm against stresses, so examining other types of structures to see how they behave in comparison is essential for improvement.

